An Empirical Study of Retrieval-Augmented Generation for Domain-Specific Question Answering

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm to ground large language models (LLMs) in external knowledge and reduce hallucinations. In this paper, we present an empirical study of RAG for domain-specific question answering in the fields of medicine and law. We compare dense retrieval (DPR, E5), sparse retrieval (BM25), and hybrid approaches combined with three LLMs: LLaMA-3-8B, Mistral-7B, and GPT-4o-mini. Experiments on 5,000 domain questions show that hybrid retrieval with reranking improves answer faithfulness by 23.4% over closed-book baselines. We further analyze chunking strategies and find that 512-token chunks with 64-token overlap achieve the best trade-off between recall and precision. Our findings provide practical guidelines for building production RAG systems.

Keywords: Retrieval-Augmented Generation, Large Language Models, Dense Retrieval, Question Answering, Hallucination

1. Introduction

Large language models have demonstrated remarkable capabilities in open-domain question answering. However, when deployed in high-stakes domains such as medicine and law, they frequently hallucinate facts, cite non-existent statutes, or provide outdated clinical guidance.

Retrieval-Augmented Generation, first proposed by Lewis et al. [1], addresses this limitation by retrieving relevant documents from an external corpus and conditioning generation on them. Despite its popularity, practitioners still face open questions: which retriever works best for Vietnamese-English mixed corpora, what chunk size is optimal, and how much does reranking help?

In this work we make three contributions: (1) a systematic comparison of retrievers and generators on two domain-specific datasets, MedQA-VN and LegalQA-VN; (2) an ablation study on chunking and top-k parameters; (3) a set of reproducible guidelines and an open-source BasicRAG pipeline.

2. Related Work

2.1 Retrieval for QA. Sparse methods such as TF-IDF and BM25 [2] remain strong baselines due to exact lexical matching. Dense passage retrieval [3] uses bi-encoders to capture semantics and generally outperforms BM25 on natural questions. Recent embedding models such as E5 [4] and BGE achieve state-of-the-art on MTEB benchmarks.

2.2 Generation with Citations. Recent work focuses on faithful generation with citations [5]. Methods include constrained decoding, post-hoc verification, and iterative retrieval (FLARE, Self-RAG). Our study builds on the standard single-hop RAG pipeline to isolate the effect of retrieval quality.

3. Methodology

Our BasicRAG pipeline consists of four stages: document ingestion, chunking, indexing, and query-time retrieval-generation. PDFs are parsed with PyMuPDF, split into overlapping chunks, embedded with E5-large, and stored in FAISS. At query time, we retrieve top-k=5 passages, optionally rerank with Cohere Rerank, and prompt the LLM with a strict grounded template.

Model	Retriever	MedQA-VN (F1)	LegalQA-VN (F1)
LLaMA-3-8B	BM25	61.2	58.7
LLaMA-3-8B	E5-large	68.4	66.1
LLaMA-3-8B	Hybrid + Rerank	72.9	71.3
GPT-4o-mini	Hybrid + Rerank	78.5	76.8

The prompt instructs the model to answer only from the provided context and to output 'I do not know' when evidence is insufficient. Temperature is set to 0.1 to reduce creativity. Evaluation metrics include Exact Match (EM), token-level F1, faithfulness (as judged by GPT-4), and citation precision.

4. Experiments

4.1 Datasets. MedQA-VN contains 2,500 Vietnamese medical exam questions paired with Ministry of Health guidelines. LegalQA-VN contains 2,500 questions about the Vietnamese Civil Code 2015 and related decrees. Each question is annotated with gold supporting passages.

4.2 Main Results. Table 1 summarizes the results. Hybrid retrieval consistently outperforms single retrievers by 4-6 F1 points. Reranking adds another 2.1 points on average. GPT-4o-mini achieves the highest absolute scores, but the open-source LLaMA-3-8B with hybrid retrieval closes 85% of the gap, making it a cost-effective alternative.

4.3 Chunking Ablation. We tested chunk sizes of 256, 512, and 1024 tokens. Size 512 with 64-token overlap yields the best recall@5 of 0.87, while 1024 introduces noise and 256 fragments legal articles. Increasing top-k beyond 5 improves recall marginally but degrades faithfulness due to distracting context, confirming the 'lost in the middle' phenomenon [6].

5. Conclusion

We presented an empirical study showing that hybrid retrieval plus reranking significantly improves domain-specific QA. Future work will extend BasicRAG to multi-hop reasoning and multimodal PDFs containing tables and scanned images. All code, datasets splits, and hyperparameters are released to facilitate reproducibility.

References

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